

## Rest and Rest Seats in RPD

### Learning Objectives

After this lecture, students will be able to:

- Understand rests and rest seats in removable partial dentures: their functions, types, and preparation procedure.
- Distinguish between different forms of rests (occlusal, lingual, incisal), and indications/contraindications.

**Rest** is a rigid extension of a partial denture that rests on a tooth's occlusal, incisal, and lingual surfaces to provide vertical support for the removable partial denture and transmit functional forces to the teeth.

**The Rest Seat** is that portion of the tooth selected and prepared to receive a rest.



### Functions of Rests

1. To direct forces along the long axis of the abutment tooth.
2. To prevent the denture base from moving cervically and impinging on the gingival tissue.
3. To serve as an indirect retainer by preventing rotation of the partial denture (Class I or II RPDs).

### **Types of rests:**

1. Occlusal rest: A rest placed on the occlusal surface of a bicuspid or molar
  2. Lingual rest: A rest placed on the cingulum of an anterior tooth (usually the canine).
  3. Incisal rest: A rest placed on an anterior tooth at the incisal edge.
- 

### **Preparation of Rest Seats**

Rest seats should be prepared using light pressure with a high-speed handpiece. Since minimal preparation is usually performed, minimal heat is generated.

Because preparations are usually entirely in enamel, it is best to avoid anaesthesia so the patient can inform the dentist when sensitivity is felt.

Occlusal rest seats can be prepared with medium round burs. Guiding planes and cingulum rest seats can be prepared with a long, medium-diameter cylindrical bur.



### **Rest Seat Form**

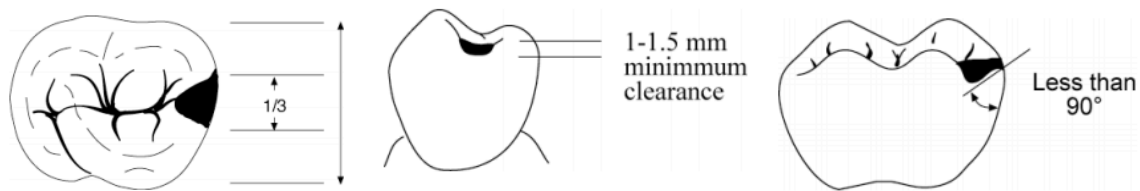
Rest seats should have a smooth-flowing outline form (i.e. no sharp line angles).

#### **1. Occlusal Rest Seats:**

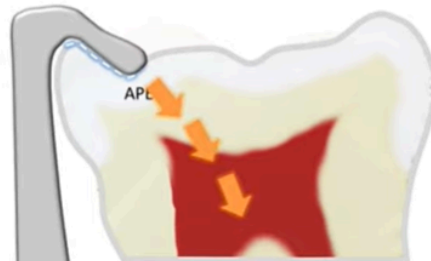
The outline of an occlusal rest seat is a rounded triangular shape with its apex nearest to the centre of the tooth.

The base of the triangular shape is at the marginal ridge and should be approximately one-third the bucco-lingual width of the tooth.

The marginal ridge must be lowered and rounded to provide sufficient metal bulk to prevent the remainder of the minor connector from fracturing (1 to 1.5 mm).



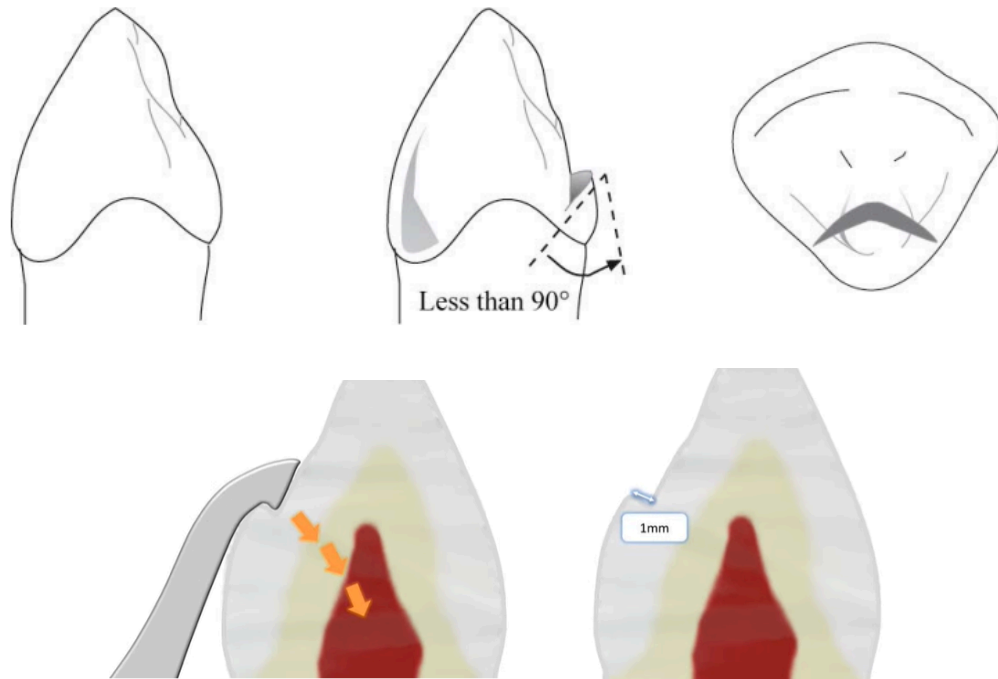
The floor of the rest seat should be inclined towards the centre of the tooth so that the angle formed by the rest and the minor connector is less than 90°. This helps direct the occlusal forces along the tooth's long axis.



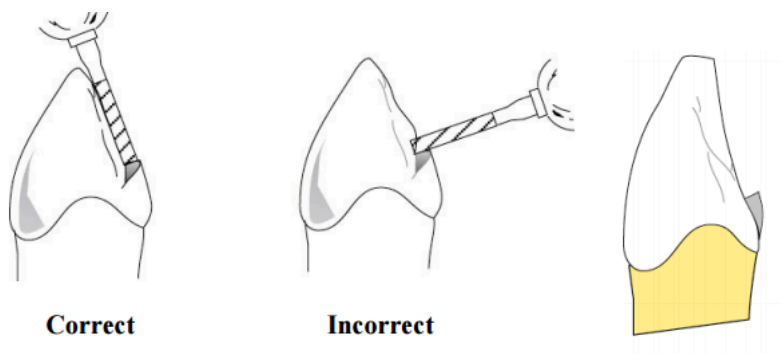
## 2. Lingual Rest Seats

Lingual or cingulum rests on anterior teeth are often utilised when no posterior teeth are present or indirect retention is necessary. The anterior tooth most readily adaptable to a cingulum rest is the canine, due to its well-developed cingulum. When a canine is not available, the cingulum of an incisor may be used. In some instances, multiple rests spread over the cingula of several teeth may be required to minimise stress on the teeth. Root form, root length, inclination of the tooth, and the crown-root ratio must be considered in the planning for using such rests.

From the *lingual aspect*, the rest seat assumes the form of a broad inverted "V" maintaining the natural contour often seen in the canine cingulum. From the *incisal view*, the rest seat is widest at the central aspect of the canine (approximately 1 mm). The *proximal view* demonstrates the angulation of the floor of the rest seat ( $< 90^\circ$ ). The borders of the rest of the seat are slightly rounded to avoid sharp line angles during its preparation.



A medium or large-diameter cylindrical fissure bur should be utilised to approach the long axis of the tooth. Approach from a horizontal direction will often result in an undercut incisal to the rest seat.



When a cingulum is poorly developed and lacks sufficient bulk to support a cingulum rest seat, a rest seat can be made using composite resin (bonded rest seats).

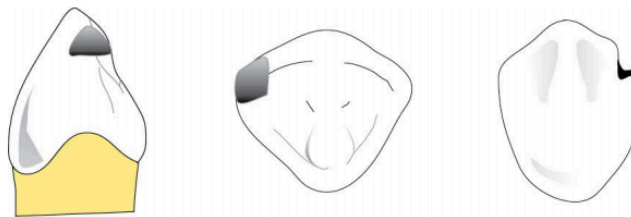
### 3. Incisal Rests seats

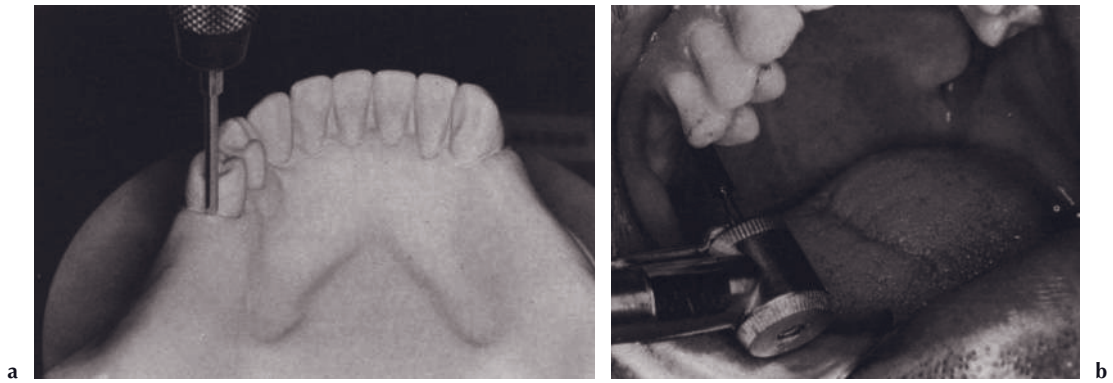
Incisal rests are inferior to lingual rests both mechanically and esthetically. Normally, they should not be used unless it is impossible to place a lingual or composite bonded rest seat.

An incisal rest seat is usually placed on the mesio- or disto-incisal angle of the incisor teeth with the deepest portion towards the tooth's centre. It is predominantly used as an auxiliary rest or an indirect retainer.

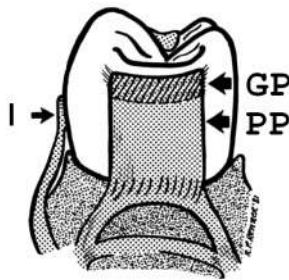
It is usually used on the mandibular incisor, where the lower lip covers as much as possible of the metal of the rest that shows at the incisal edge.

Mechanically, a lingual rest is preferable to an incisal rest, because the lingual rest is placed nearer to the centre of rotation of the tooth and therefore, will have less tendency to tip the tooth.

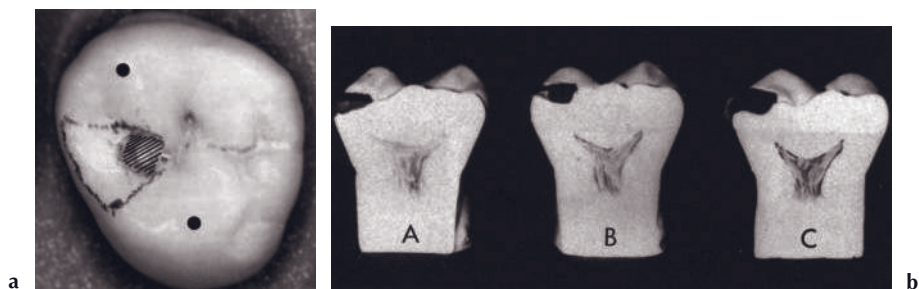




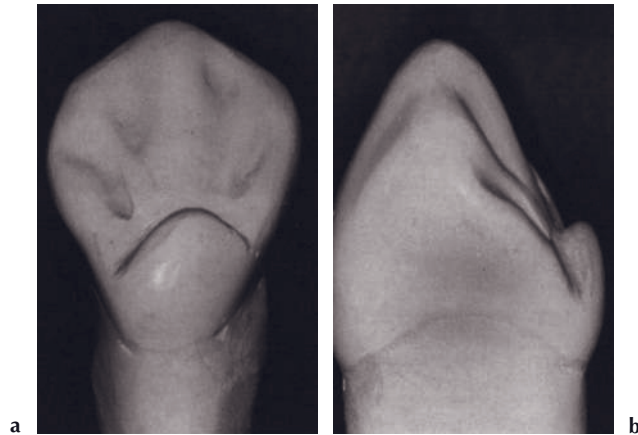
**Figures 4.1.1a and b.** The guide plane is parallel to the path of insertion, shown here using a surveyor with analyzing rod to demonstrate the path of insertion/removal (a). The guide plane is created using parallel sided burs, intraorally (b).



**Figure 4.1.2.** The length of the guide plane varies according to design philosophy; however, the guide plane is prepared on the interproximal surface, line angle to line angle from a buccolingual perspective. The guide plane (GP) is on the tooth surface, the proximal plate (PP) is the metal component of the RPD framework, and an I-bar (I) is shown on the buccal surface of this diagram.



**Figure 4.1.3.** a. Occlusal rest seat area is shown from an occlusal view. The typical occlusal rest is a spoon-shaped, rounded triangle. It should be 1.5 mm thick occlusogingivally, at the junction of the rest seat and minor connector. It is one-third the width of the tooth buccolingually. b. Three examples of occlusal rest seat preparation in cross-section. (A) and (B) illustrate poor rest seat preparations that do not provide a positive seat for a metal rest. (B) demonstrates inadequate marginal ridge reduction, leaving a sharp edge on the ridge. (C) shows an acceptable form for an occlusal rest seat.



**Figures 4.1.4a and b.** The cingulum rest seat preparation requires a prominent cingulum. The preparation is a chevron or inverted “V”-shape mesiodistally (a), and a concave “V”-shape buccolingually (b). The internal line angles should be rounded. When an anterior tooth does not have a cingulum that is prominently sufficient to prepare without exposing the dentin, composite resin may be used to develop the prominence needed; the cingulum rest should be created such that at least half of the rest seat is in enamel. The careful use of composite resin on a cingulum, either mandibular or maxillary canine, is preferable to preparing an incisal rest seat since use of an incisal rest seat creates an unacceptable esthetic result.